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EEE 554 Lecture #11

Covariance Matrix

The covariance matrix of a random vector is the $n \times n$ matrix where

$$\begin{aligned} \underline{C}_X &= E[(\underline{X} - M_X)(\underline{X} - M_X)^T] \\ &= \begin{bmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_n) \\ \vdots & \vdots & & \vdots \\ \text{cov}(X_n, X_1) & \text{cov}(X_n, X_2) & \dots & \text{var}(X_n) \end{bmatrix} \\ M_X &= \begin{pmatrix} \mu_{X_1} \\ \vdots \\ \mu_{X_n} \end{pmatrix} = E \begin{bmatrix} X \\ \vdots \end{bmatrix} \end{aligned}$$

Note that $\text{cov}(\underline{x}_i, \underline{x}_j) = \text{cov}(\underline{x}_j, \underline{x}_i)$ implies C_x is symmetric. Because C_x has the form $\underline{A}\underline{A}^T$, it is non-negative definite.

Example: Suppose

$$C_X = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}, M_X = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

which has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = 2$ and corresponding eigenvectors:

$$E_1 = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, E_2 = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Plotting these eigenvectors weighted by their corresponding eigenvalues, centered at M_x , gives the axes of the covariance Ellipse. Note the angle of the ellipse of these negatively correlated variables.

Note that $\underline{C}_X = E[\underline{X}\underline{X}^T] - M_X M_X^T$

Linear Transformations

If A is a (deterministic) $m \times n$ matrix and $\underline{X}$ a random n -vector, then $\underline{Y} = \underline{A}\underline{X}$ is a random m -vector. Recall that

$$E[\underline{Y}] = E[\underline{A}\underline{X}] = \underline{A}E[\underline{X}]$$

The covariance matrix of $\underline{Y}$ is

$$\begin{aligned} C_Y &= E[(\underline{Y} - M_Y)(\underline{Y} - M_Y)^T] \\ &= E[(\underline{A}\underline{X} - \underline{A}M_X)(\underline{A}\underline{X} - \underline{A}M_X)^T] \\ &= \underline{A}E[(\underline{X} - M_X)(\underline{X} - M_X)^T]\underline{A}^T \\ &= \underline{A}C_X \underline{A}^T \end{aligned}$$

Multivariate Normal Distribution

A random vector is multivariate normal (Gaussian) if its PDF

$$f_x(x) = \frac{1}{(2\pi)^{n/2} \det(C)^{1/2}} \exp \left\{ -\frac{1}{2} (\underline{X} - m)^T C^{-1} (\underline{X} - m) \right\}$$

for a positive definite matrix C and a deterministic vector M .

Notes: * Direct integration confirms $\int_{\mathbb{R}^n} f_{\underline{X}}(\underline{X}) dX = 1$. * Application of the expectation rule shows C : * $E[\underline{X}] = \int_{\mathbb{R}^n} \underline{X} f_{\underline{X}}(\underline{X}) dX = M$ * $C_x = \int_{\mathbb{R}^n} (\underline{X} - M)(\underline{X} - M)^T f_{\underline{X}}(\underline{X}) dX = C$

With parameters M and C , one denotes $\underline{X} \sim N[M, C]$.

Uncorrelated Normal Random Vectors

a A random vector $\underline{X}$ is uncorrelated if C_x is a diagonal matrix. If $\underline{X}$ is Gaussian and uncorrelated, then

$$C_X = \begin{bmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_n^2 \end{bmatrix}$$

and $\det C_X = \sigma_1^2 \dots \sigma_n^2$.

So

$$\begin{aligned} f_{\underline{X}}(\underline{X}) &= \frac{1}{\sqrt{(2\pi)^n \sigma_1^2 \dots \sigma_n^2}} \exp \left\{ -\frac{1}{2} \sum_{k=1}^n (x_k - \mu_k)^2 \sigma_k^{-2} \right\} \\ &= \frac{1}{\sqrt{2\pi \sigma_1^2}} \exp \left\{ -\frac{(X_1 - \mu_1)^2}{2\sigma_1^2} \right\} \dots \frac{1}{\sqrt{2\pi \sigma_n^2}} \exp \left\{ -\frac{(X_n - \mu_n)^2}{2\sigma_n^2} \right\} \\ &= f_{x_1}(x_1) \dots f_{x_n}(x_n) \end{aligned}$$

i.e., the components of an uncorrelated Gaussian random vector are independent Gaussian RVs.

Decorrelation and Whitening

Suppose $u_1, \dots, u_n$ are orthonormal eigenvectors of C_x . Then the matrix

$$u^T = [u_1 \dots u_n]$$

is unitary and the random vector $y = ux$ has covariance matrix

$$\begin{aligned} C_y &= u C_x u^T = u [\lambda_1 u_1 \dots \lambda_n u_n] \\ &= \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \end{aligned}$$

where $\lambda_1, \dots, \lambda_n$ are the (non-negative) eigenvalues corresponding to $u_1, \dots, u_n$ respectively.

Note that Y is an uncorrelated random vector. A random vector is white if C_y is a diagonal matrix; under some circumstances one also requires $721, M$. With the notation above, define

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}} u_1 & \dots & \frac{1}{\sqrt{\lambda_n}} u_n \end{bmatrix}$$

If $y = VX$, then

$$C_y = V C_x V^T = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = I$$

so Y is white.